

Emissions Over Time All

Zhang Zhenglin

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Necessary libraries

```
library(knitr)
library(ggplot2)
theme_set(theme_bw())
library(emmeans)
library(multcomp)
library(PLS205)
library(lme4)
library(lmerTest)
library(multcompView)
library(car)
library(Rmisc)
library(dplyr) #https://r4ds.had.co.nz/ (Chapter 3, Chapter 5, look at filter and select)
```

```
# https://bookdown.org/ansellbr/WEHI\_tidyR\_course\_book/
library(stringr)
library(data.table)
library(GGally)
library(formatR)
library(readxl)
library(ggpubr)
library(lubridate)
library(paletteer)
library(forcats)
```

Data Organisation

Read from excel

```
#read directly from the directory where I generated the files from
master <- read_excel("D:/Academics/UC Davis/School Work/Linquist Lab/Data/R stats/GHG and MAOM POM/Daily")
str(master)
```

```
## tibble [746 x 4] (S3: tbl_df/tbl/data.frame)
## $ Plot      : chr [1:746] "106" "107" "204" "209" ...
## $ Date      : POSIXct[1:746], format: "2021-05-11" "2021-05-11" ...
## $ CH4_g_ha_day: num [1:746] 0 0 0 0 0 ...
## $ N2O_g_ha_day: num [1:746] 0 0 0 0 0 ...
```

2021/2022

```
CRFRF_2021 <- master %>%
  filter(Date >= as.POSIXct("2021-05-10") & Date <= as.POSIXct("2022-05-12"))%>%
  mutate(Treatment = case_when(
    Plot %in% c("106", "204", "302") ~ "FR",
    Plot %in% c("107", "209", "307") ~ "CR",
    Plot %in% c("402", "505", "601") ~ "F",
    TRUE ~ "Other" # This line handles cases where plot is not listed
  ))
```

```
CRFRF_2021 %>%
  group_by(Date, Treatment) %>%
  mutate(CH4_g_ha_day_se = sd(CH4_g_ha_day)/sqrt(3)) %>%
  summarise(CH4_g_ha_day = mean(CH4_g_ha_day),
            CH4_g_ha_day_se = mean(CH4_g_ha_day_se)) %>%
  ungroup() %>%
  group_by(Treatment)%>%
  summarise(max(CH4_g_ha_day))
```

```
## 'summarise()' has grouped output by 'Date'. You can override using the
## '.groups' argument.
```

```
## # A tibble: 3 x 2
##   Treatment 'max(CH4_g_ha_day)'
##   <chr>          <dbl>
## 1 CR              7731.
## 2 F                4.89
## 3 FR             4628.
```

2021/22 CH4

```
CH4_g_ha_day_2021_graphing <- CRFRF_2021 %>% group_by(Date, Treatment) %>%
mutate(CH4_g_ha_day_se = sd(CH4_g_ha_day)/sqrt(3)) %>%
  summarise(CH4_g_ha_day = mean(CH4_g_ha_day),
            CH4_g_ha_day_se = mean(CH4_g_ha_day_se))
```

```
## 'summarise()' has grouped output by 'Date'. You can override using the
## '.groups' argument.
```

```
CH4_2021_graph <-
  ggplot(CH4_g_ha_day_2021_graphing, aes(x= Date, y = CH4_g_ha_day, color = Treatment))+
  geom_point()+
  geom_line()+
  geom_errorbar(aes(ymin=CH4_g_ha_day-CH4_g_ha_day_se, ymax=CH4_g_ha_day+CH4_g_ha_day_se))+
  scale_color_manual(values = c("CR" = "#0072B2", "FR" = "#FFCC66", "F" = "black"),
                    labels = c(expression("Continuous Rice (CR)"),
                               expression("Fallow (F)"),
                               expression("Fallow Rice (FR)"))
                    )
                    )+
  labs(y = expression("Daily CH"[4]~"Flux (g ha"^-1~" day"^-1~")")) +
  scale_y_continuous(limits = c(-2000, 14000),
                    expand = c(0, 0),
                    breaks = seq(0, 14000, by = 2500))+
  scale_x_datetime(name = "",
                  limits = as.POSIXct(c("2021-05-01", "2022-05-15")), # Set x-axis limits
                  date_labels = "%b %Y",
                  date_breaks = "1 month")+
  ggtitle("2021/22") +
  theme_classic()+
  theme(legend.title = element_blank(),
        axis.text.x = element_text(size=10,angle=50, hjust=1.2, vjust=1.2))+
  geom_vline(xintercept = as.POSIXct(c("2021-10-12")), linetype = "dashed", color = "black", size = 0.4)+
  annotate("point", x = as.POSIXct(c("2021-05-13",
                                     "2021-11-23")), y = -1300, shape = 25, color = "#017d35", fill = "white",
            size = 100, stroke = "black")+
  annotate("point", x = as.POSIXct(c("2021-09-05",
                                     "2022-02-04")), y = -1300, shape = 25, color = "#017d35", stroke = "black",
            size = 100, fill = "white")+
  annotate(
    "text",
    x = as.POSIXct(c("2021-07-15")), # X-axis positions for annotations
    y = c(12500), # Y-axis positions for annotations
    label = expression("Summer"),
    size = 4,
```

```

vjust = 0 # Adjust vertical position of asterisks
)+
  annotate(
    "text",
    x = as.POSIXct(c("2022-02-10")), # X-axis positions for annotations
    y = c(12500), # Y-axis positions for annotations
    label = expression("Winter"),
    size = 4,
    vjust = 0 # Adjust vertical position of asterisks
  )

```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

CH4_2021_graph

```

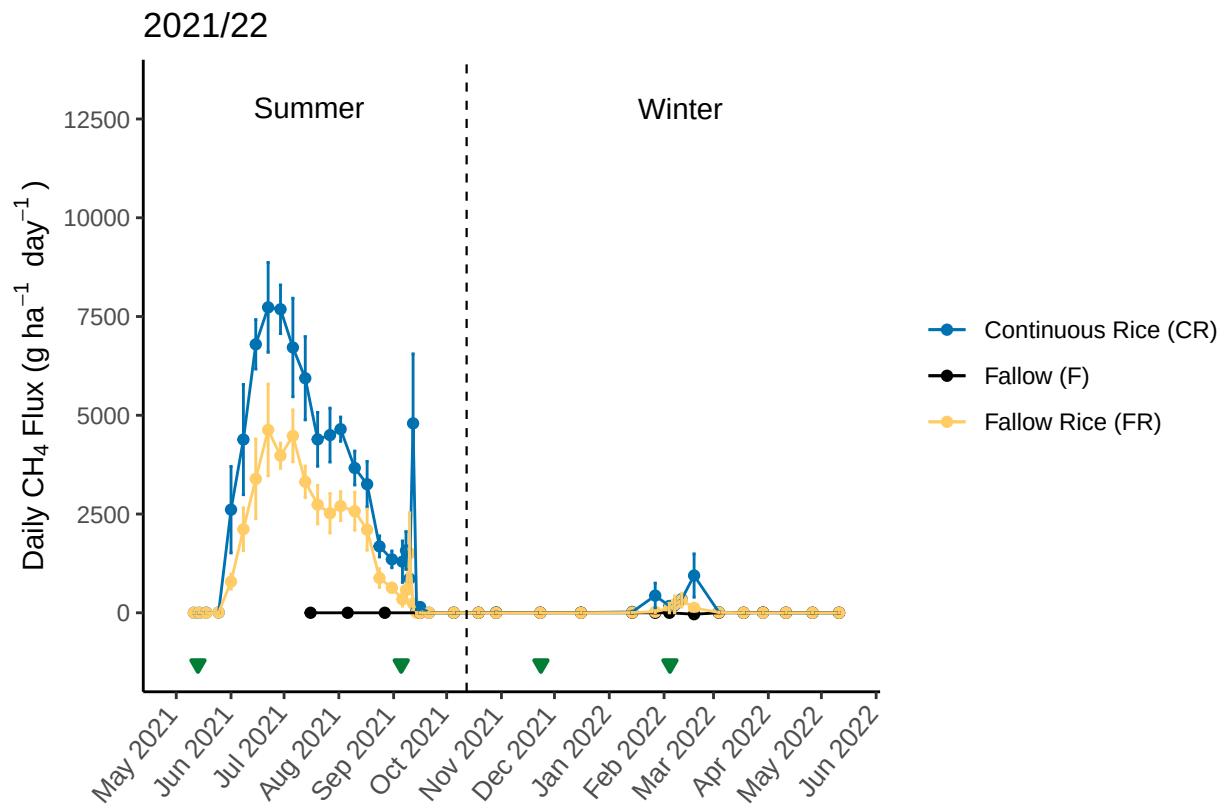
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```

```

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```



2021/22 N2O

```
N2O_g_ha_day_2021_graphing <- CRFRF_2021 %>% group_by(Date, Treatment) %>%
mutate(N2O_g_ha_day_se = sd(N2O_g_ha_day)/sqrt(3)) %>%
  summarise(N2O_g_ha_day = mean(N2O_g_ha_day),
            N2O_g_ha_day_se = mean(N2O_g_ha_day_se))
```

'summarise()' has grouped output by 'Date'. You can override using the
'.groups' argument.

```
N2O_2021_graph <-
  ggplot(N2O_g_ha_day_2021_graphing, aes(x= Date, y = N2O_g_ha_day, color = Treatment))+
  geom_point()+
  geom_line()+
  geom_errorbar(aes(ymin=N2O_g_ha_day-N2O_g_ha_day_se, ymax=N2O_g_ha_day+N2O_g_ha_day_se))+
  scale_color_manual(values = c("CR" = "#0072B2", "FR" = "#FFCC66", "F" = "black"),
                    labels = c(expression("Continuous Rice (CR)"),
                                expression("Fallow (F)"),
                                expression("Fallow Rice (FR)"))
                    )
                    )+
  labs(y = expression("Daily N"[2]~"0 Flux (g ha"^-1~" day"^-1~"")) +
  scale_y_continuous(limits = c(-70,500),
                    expand = c(0, 0),
                    breaks = c(0,100,200,300,400,500)
                    )+
  scale_x_datetime(name = "",
                  limits = as.POSIXct(c("2021-05-01", "2022-05-15")), # Set x-axis limits
                  date_labels = "%b %Y",
                  date_breaks = "1 month")+
  ggtitle("2021/22") +
  theme_classic()+
  theme(legend.title = element_blank(),
        axis.text.x = element_text(size=10,angle=50, hjust=1.2, vjust=1.2))+
  geom_vline(xintercept = as.POSIXct(c("2021-10-12")), linetype = "dashed", color = "black", size =0.4)+
  annotate("point", x = as.POSIXct(c("2021-05-13",
                                     "2021-11-23")), y = -30, shape = 25, color = "#017d35", fill ="#017d35",
          annotate("point", x = as.POSIXct(c("2021-09-05",
                                     "2022-02-04")), y = -30, shape = 25, color = "#017d35", fill ="#017d35",
          annotate(
            "text",
            x = as.POSIXct(c("2021-07-15")), # X-axis positions for annotations
            y = c(400), # Y-axis positions for annotations
            label = expression("Summer"),
            size = 4,
            vjust = 0 # Adjust vertical position of asterisks
          )+
  annotate(
    "text",
    x = as.POSIXct(c("2022-02-10")), # X-axis positions for annotations
    y = c(400), # Y-axis positions for annotations
    label = expression("Winter"),
```

```

size = 4,
vjust = 0 # Adjust vertical position of asterisks
)

```

N2O_2021_graph

```

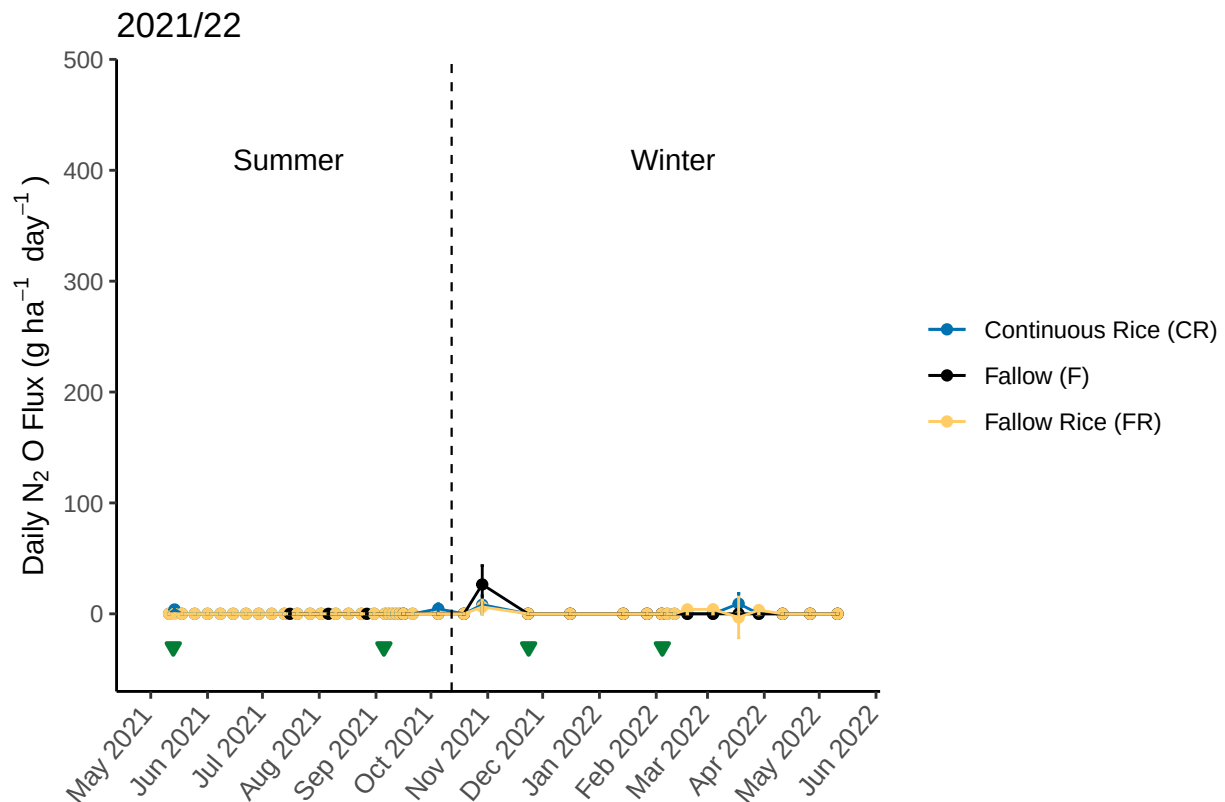
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```

```

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```



2022/23

```

CRFRF_2022 <- master %>%
  filter(Date >= as.POSIXct("2022-05-15") & Date <= as.POSIXct("2023-04-19"))%>%
  filter (Plot %in% c(402, 409, 505, 512, 601, 608, 209, 307, "K1"))%>%
  mutate(Treatment = case_when(
    Plot %in% c("402", "505", "601") ~ "FR",
    Plot %in% c("409", "512", "608") ~ "CR",
    Plot %in% c("209", "307", "K1") ~ "F",
  )

```

```

    TRUE ~ "Other" # This line handles cases where plot is not listed
  ))

CRFRF_2022 %>%
  group_by(Date, Treatment) %>%
  mutate(CH4_g_ha_day_se = sd(CH4_g_ha_day)/sqrt(3)) %>%
  summarise(CH4_g_ha_day = mean(CH4_g_ha_day),
            CH4_g_ha_day_se = mean(CH4_g_ha_day_se)) %>%
  ungroup() %>%
  group_by(Treatment)%>%
  summarise(max(CH4_g_ha_day))

```

'summarise()' has grouped output by 'Date'. You can override using the
'.groups' argument.

```

## # A tibble: 3 x 2
##   Treatment 'max(CH4_g_ha_day)'
##   <chr>          <dbl>
## 1 CR              6077.
## 2 F               15.7
## 3 FR             7558.

```

2022/2023 CH4

```

CH4_g_ha_day_2022_graphing <- CRFRF_2022 %>% group_by(Date, Treatment) %>%
mutate(CH4_g_ha_day_se = sd(CH4_g_ha_day)/sqrt(3)) %>%
  summarise(CH4_g_ha_day = mean(CH4_g_ha_day),
            CH4_g_ha_day_se = mean(CH4_g_ha_day_se))

```

'summarise()' has grouped output by 'Date'. You can override using the
'.groups' argument.

```

CH4_2022_graph <-
  ggplot(CH4_g_ha_day_2022_graphing, aes(x= Date, y = CH4_g_ha_day, color = Treatment))+
  geom_point()+
  geom_line()+
  geom_errorbar(aes(ymin=CH4_g_ha_day-CH4_g_ha_day_se, ymax=CH4_g_ha_day+CH4_g_ha_day_se))+
  scale_color_manual(values = c("CR" = "#0072B2", "FR" = "#FFCC66", "F" = "black"),
                    labels = c(expression("Continuous Rice (CR)"),
                               expression("Fallow (F)"),
                               expression("Fallow Rice (FR)"))
                    )
  )+
  labs(y = expression("Daily CH"[4]~"Flux (g ha"^-1~" day"^-1~")")) +
  scale_y_continuous(limits = c(-2000, 14000),
                    expand = c(0, 0),
                    breaks = seq(0, 14000, by = 2500))+
  scale_x_datetime(name = "",
                  limits = as.POSIXct(c("2022-05-01", "2023-05-15")), # Set x-axis limits
                  date_labels = "%b %Y",

```

```

                                date_breaks = "1 month")+
ggtitle("2022/23") +
theme_classic()+
theme(legend.title = element_blank(),
      axis.text.x = element_text(size=10,angle=50, hjust=1.2, vjust=1.2))+
geom_vline(xintercept = as.POSIXct(c("2022-10-20")), linetype = "dashed", color = "black", size =0.4)
annotate("point", x = as.POSIXct(c("2022-05-21",
                                "2022-12-20")), y = -1300, shape = 25, color = "#017d35", fill =
      annotate("point", x = as.POSIXct(c("2022-09-16",
                                "2023-02-09")), y = -1300, shape = 25, color = "#017d35", fill =

annotate(
  "text",
  x = as.POSIXct(c("2022-07-15")), # X-axis positions for annotations
  y = c(12500), # Y-axis positions for annotations
  label = expression("Summer"),
  size = 4,
  vjust = 0 # Adjust vertical position of asterisks
)+
  annotate(
    "text",
    x = as.POSIXct(c("2023-02-10")), # X-axis positions for annotations
    y = c(12500), # Y-axis positions for annotations
    label = expression("Winter"),
    size = 4,
    vjust = 0 # Adjust vertical position of asterisks
  )
CH4_2022_graph

```

```

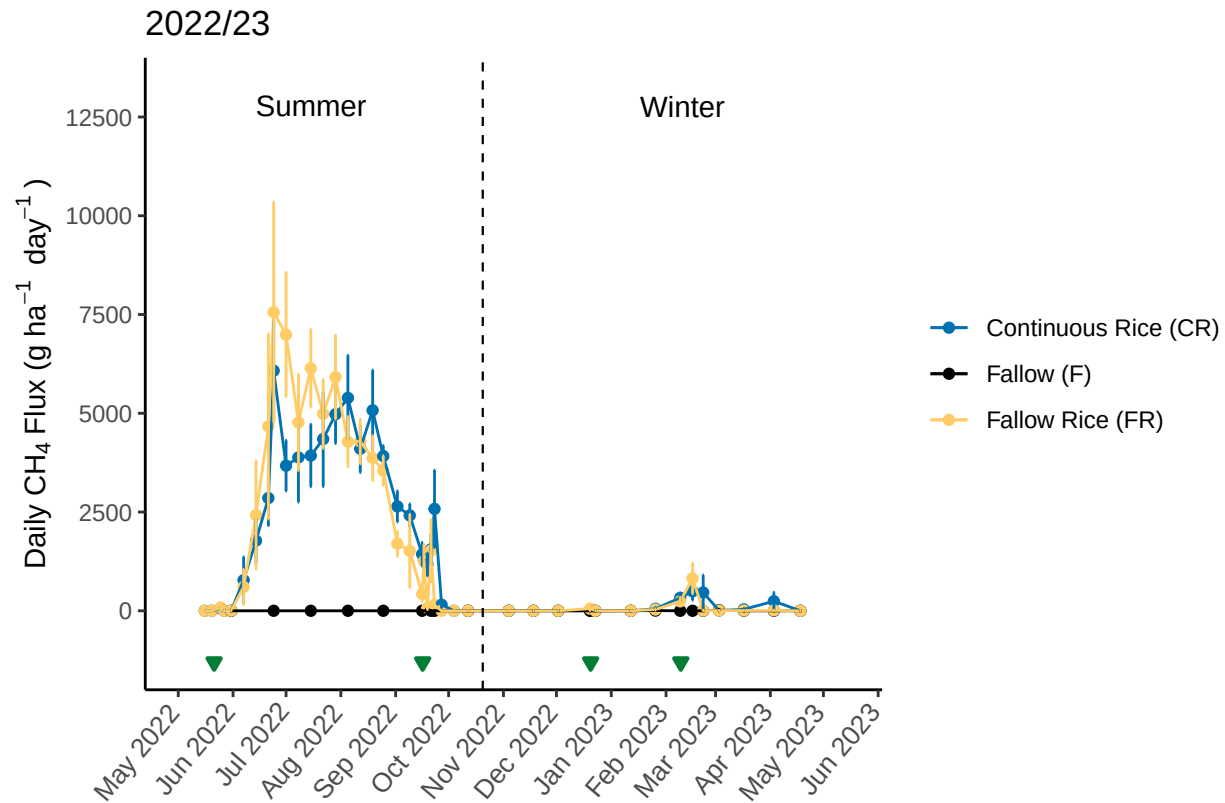
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```

```

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```



2022/2023 N₂O

```
N2O_g_ha_day_2022_graphing <- CRFRF_2022 %>% group_by(Date, Treatment) %>%
mutate(N2O_g_ha_day_se = sd(N2O_g_ha_day)/sqrt(3)) %>%
  summarise(N2O_g_ha_day = mean(N2O_g_ha_day),
            N2O_g_ha_day_se = mean(N2O_g_ha_day_se))
```

'summarise()' has grouped output by 'Date'. You can override using the
'.groups' argument.

```
N2O_2022_graph <-
  ggplot(N2O_g_ha_day_2022_graphing, aes(x= Date, y = N2O_g_ha_day, color = Treatment))+
  geom_point()+
  geom_line()+
  geom_errorbar(aes(ymin=N2O_g_ha_day-N2O_g_ha_day_se, ymax=N2O_g_ha_day+N2O_g_ha_day_se))+
  scale_color_manual(values = c("CR" = "#0072B2", "FR" = "#FFCC66", "F" = "black"),
                    labels = c(expression("Continuous Rice (CR)"),
                               expression("Fallow (F)"),
                               expression("Fallow Rice (FR)"))
  )+
  labs(y = expression("Daily N[2]~"O Flux (g ha^-1~" day^-1~")")) +
  scale_y_continuous(limits = c(-70,500),
```

```

        expand = c(0, 0),
        breaks = c(0,100,200,300,400,500)
    )+
scale_x_datetime(name = "",
                  limits = as.POSIXct(c("2022-05-01", "2023-05-15")), # Set x-axis limits
                  date_labels = "%b %Y",
                  date_breaks = "1 month")+
ggtitle("2022/23") +
theme_classic()+
theme(legend.title = element_blank(),
      axis.text.x = element_text(size=10,angle=50, hjust=1.2, vjust=1.2))+
geom_vline(xintercept = as.POSIXct(c("2022-10-20")), linetype = "dashed", color = "black", size =0.4)+
annotate("point", x = as.POSIXct(c("2022-05-21",
                                   "2022-12-20")), y = -30, shape = 25, color = "#017d35", fill = "#017d35"),
         annotate("point", x = as.POSIXct(c("2022-09-16",
                                   "2023-02-09")), y = -30, shape = 25, color = "#017d35",fill = "#017d35"),
         annotate(
           "text",
           x = as.POSIXct(c("2022-07-15")), # X-axis positions for annotations
           y = c(400), # Y-axis positions for annotations
           label = expression("Summer"),
           size = 4,
           vjust = 0 # Adjust vertical position of asterisks
         )+
         annotate(
           "text",
           x = as.POSIXct(c("2023-02-10")), # X-axis positions for annotations
           y = c(400), # Y-axis positions for annotations
           label = expression("Winter"),
           size = 4,
           vjust = 0 # Adjust vertical position of asterisks
         )
)

N20_2022_graph

```

```

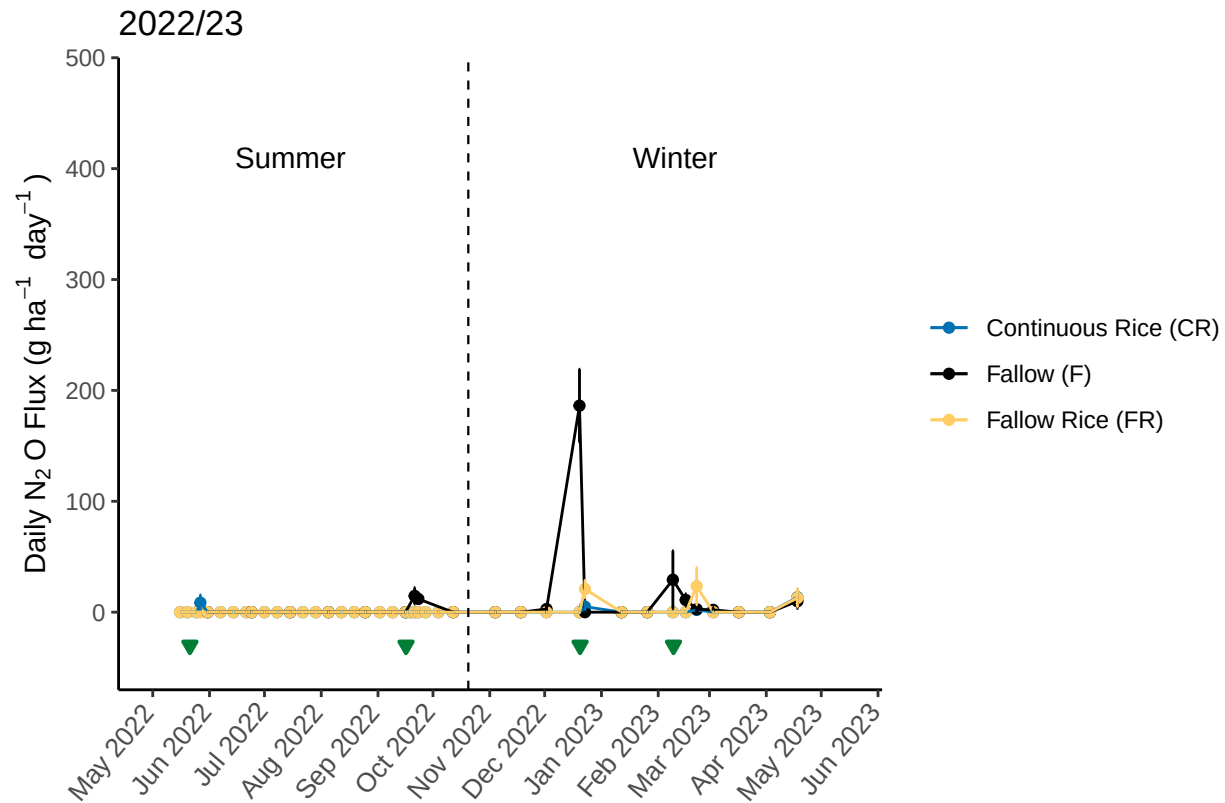
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```

```

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```



2023

```
CRFRF_2023 <- master %>%
  filter(Date >= as.POSIXct("2023-05-21") & Date <= as.POSIXct("2023-10-2")) %>%
  mutate(Treatment = case_when(
    Plot %in% c("701", "805", "903") ~ "FR",
    Plot %in% c("711", "812", "909") ~ "CR",
    TRUE ~ "Other" # This line handles cases where plot is not listed
  ))
```

```
CRFRF_2023 %>%
  group_by(Date, Treatment) %>%
  mutate(CH4_g_ha_day_se = sd(CH4_g_ha_day)/sqrt(3)) %>%
  summarise(CH4_g_ha_day = mean(CH4_g_ha_day),
            CH4_g_ha_day_se = mean(CH4_g_ha_day_se)) %>%
  ungroup() %>%
  group_by(Treatment) %>%
  summarise(max(CH4_g_ha_day))
```

```
## 'summarise()' has grouped output by 'Date'. You can override using the
## '.groups' argument.
```

```
## # A tibble: 2 x 2
```

```
## Treatment 'max(CH4_g_ha_day)'
## <chr> <dbl>
## 1 CR 8207.
## 2 FR 4416.
```

2023 CH4

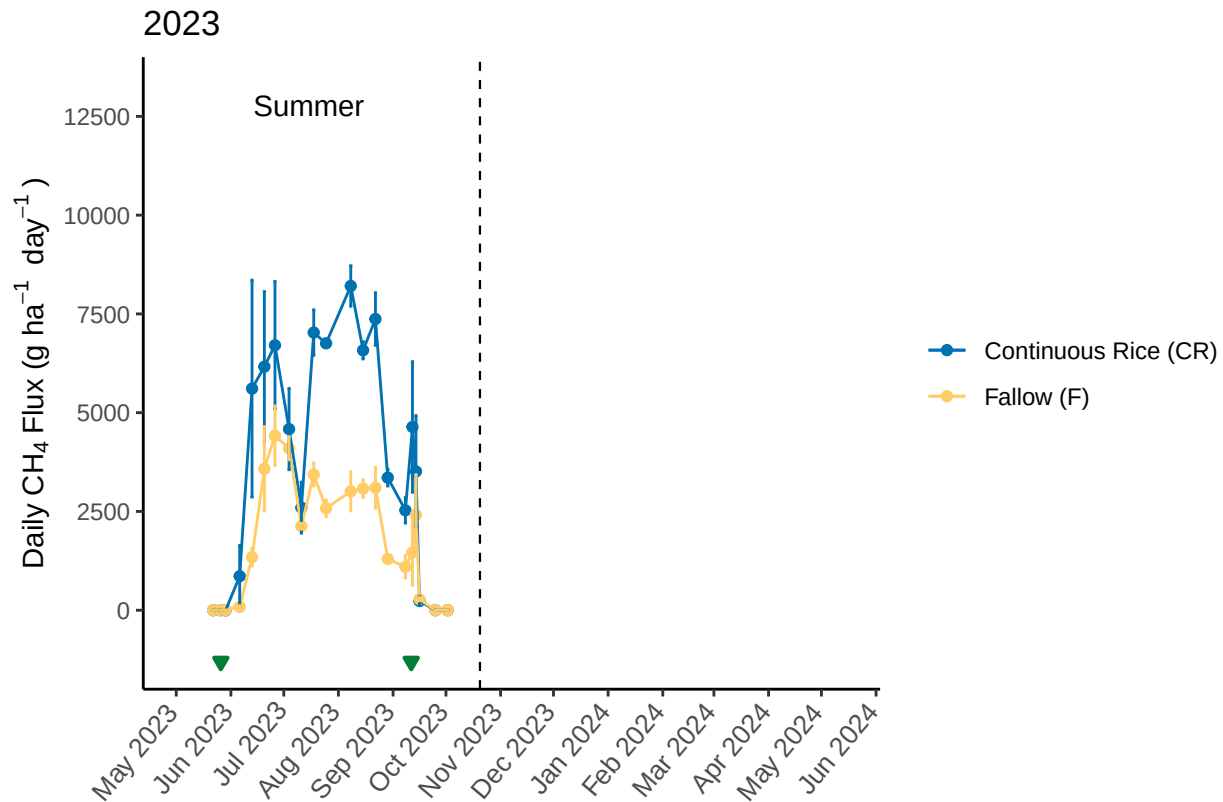
```
CH4_g_ha_day_2023_graphing <- CRFRF_2023 %>% group_by(Date, Treatment) %>%
mutate(CH4_g_ha_day_se = sd(CH4_g_ha_day)/sqrt(3)) %>%
  summarise(CH4_g_ha_day = mean(CH4_g_ha_day),
            CH4_g_ha_day_se = mean(CH4_g_ha_day_se))
```

```
## 'summarise()' has grouped output by 'Date'. You can override using the
## '.groups' argument.
```

```
CH4_2023_graph <-
  ggplot(CH4_g_ha_day_2023_graphing, aes(x= Date, y = CH4_g_ha_day, color = Treatment))+
  geom_point()+
  geom_line()+
  geom_errorbar(aes(ymin=CH4_g_ha_day-CH4_g_ha_day_se, ymax=CH4_g_ha_day+CH4_g_ha_day_se))+
  scale_color_manual(values = c("CR" = "#0072B2", "FR" = "#FFCC66", "F" = "black"),
                    labels = c(expression("Continuous Rice (CR)"),
                               expression("Fallow (F)"),
                               expression("Fallow Rice (FR)"))
                    )
  )+
  labs(y = expression("Daily CH"[4]~"Flux (g ha"^-1~" day"^-1~")")) +
  scale_y_continuous(limits = c(-2000, 14000),
                    expand = c(0, 0),
                    breaks = seq(0, 14000, by = 2500))+
  scale_x_datetime(name = "",
                  limits = as.POSIXct(c("2023-05-01", "2024-05-15")), # Set x-axis limits
                  date_labels = "%b %Y",
                  date_breaks = "1 month")+
  ggtitle("2023") +
  theme_classic()+
  theme(legend.title = element_blank(),
        axis.text.x = element_text(size=10,angle=50, hjust=1.2, vjust=1.2))+
  geom_vline(xintercept = as.POSIXct(c("2023-10-20")), linetype = "dashed", color = "black", size = 0.4)+
  annotate("point", x = as.POSIXct(c("2023-05-26")), y = -1300, shape = 25, color = "#017d35", fill = "#017d35")+
  annotate("point", x = as.POSIXct(c("2023-09-11")), y = -1300, shape = 25, color = "#017d35", fill = "#017d35")+
  annotate(
    "text",
    x = as.POSIXct(c("2023-07-15")), # X-axis positions for annotations
    y = c(12500), # Y-axis positions for annotations
    label = expression("Summer"),
    size = 4,
    vjust = 0 # Adjust vertical position of asterisks
  )
```

```
CH4_2023_graph
```

```
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'
```



2023 N2O

```
N2O_g_ha_day_2023_graphing <- CRFRF_2023 %>% group_by(Date, Treatment) %>%
mutate(N2O_g_ha_day_se = sd(N2O_g_ha_day)/sqrt(3)) %>%
  summarise(N2O_g_ha_day = mean(N2O_g_ha_day),
            N2O_g_ha_day_se = mean(N2O_g_ha_day_se))
```

```
## 'summarise()' has grouped output by 'Date'. You can override using the
## '.groups' argument.
```

```
N2O_2023_graph <-
  ggplot(N2O_g_ha_day_2023_graphing, aes(x= Date, y = N2O_g_ha_day, color = Treatment))+
  geom_point()+
  geom_line()+
  geom_errorbar(aes(ymin=N2O_g_ha_day-N2O_g_ha_day_se, ymax=N2O_g_ha_day+N2O_g_ha_day_se))+
  scale_color_manual(values = c("CR" = "#0072B2", "FR" = "#FFCC66", "F" = "black"),
                    labels = c(expression("Continuous Rice (CR)"),
                                expression("Fallow (F)"),
                                expression("Fallow Rice (FR)"))
  )
```

```

    )+
labs(y = expression("Daily N"[2]~"0 Flux (g ha"^-1~" day"^-1~"")) +
scale_y_continuous(limits = c(-70,500),
                    expand = c(0, 0),
                    breaks = c(0,100,200,300,400,500)
                    )+
scale_x_datetime(name = "",
                 limits = as.POSIXct(c("2023-05-01", "2024-05-15")), # Set x-axis limits
                 date_labels = "%b %Y",
                 date_breaks = "1 month")+
ggtitle("2023") +
theme_classic()+
theme(legend.title = element_blank(),
      axis.text.x = element_text(size=10,angle=50, hjust=1.2, vjust=1.2))+
geom_vline(xintercept = as.POSIXct(c("2023-10-20")), linetype = "dashed", color = "black", size = 0.4)
annotate("point", x = as.POSIXct(c("2023-05-26")), y = -30, shape = 25, color = "#017d35", fill = "#017d35")
  annotate("point", x = as.POSIXct(c("2023-09-11")), y = -30, shape = 25, color = "#017d35", fill = "#017d35")
  annotate(
    "text",
    x = as.POSIXct(c("2023-07-15")), # X-axis positions for annotations
    y = c(400), # Y-axis positions for annotations
    label = expression("Summer"),
    size = 4,
    vjust = 0 # Adjust vertical position of asterisks
  )
)

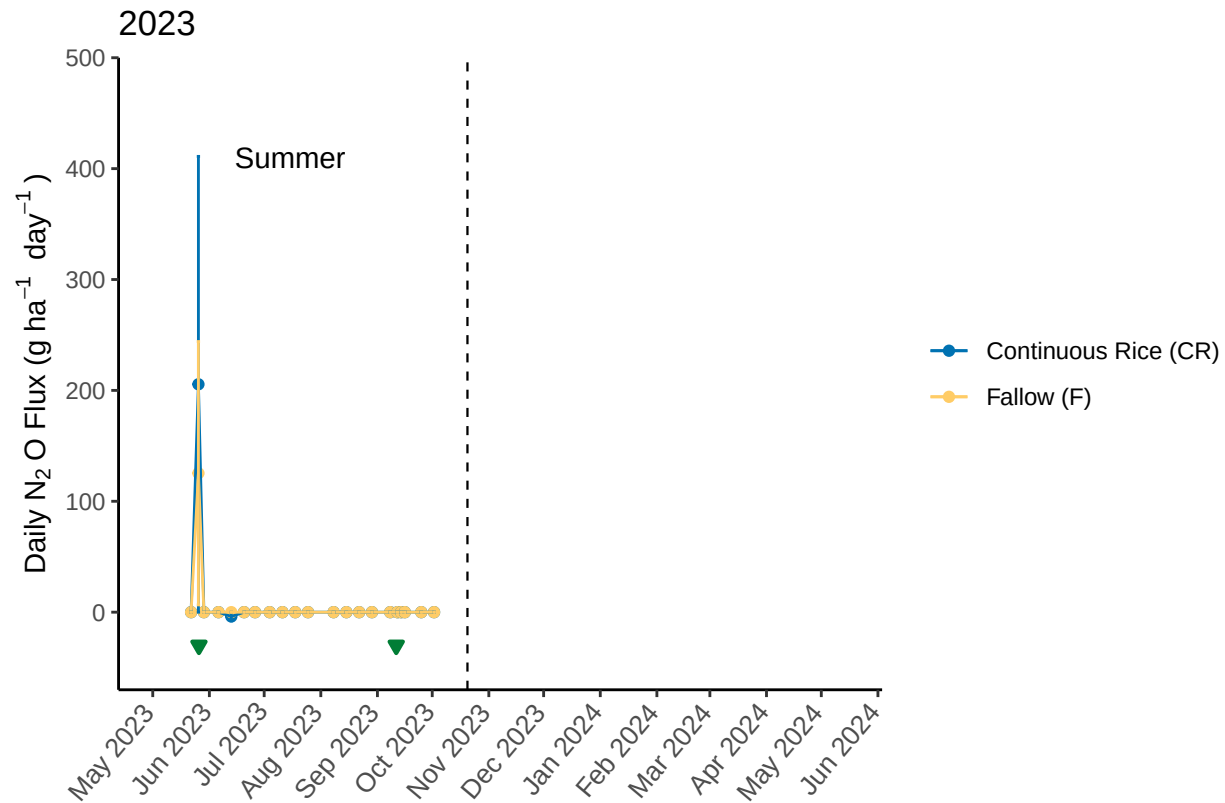
```

N2O_2023_graph

```

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

```



Combine CR/FRsummer_winter into one figure

```
all_years_CH4 <- ggarrange(CH4_2021_graph, CH4_2022_graph, CH4_2023_graph,
  nrow = 3,
  common.legend = TRUE,
  legend= "bottom")
```

```
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'
```

```
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'
```

```
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'
```

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## 'expression'
```

```
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'
```

```
## Warning in is.na(x): is.na() applied to non-(list or vector) of type
```

```
## 'expression'

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

ggsave(all_years_CH4, filename = "all_years_CH4.jpg", height = 25, width = 25, units = "cm", dpi=400)

all_years_N2O <- ggarrange(N2O_2021_graph, N2O_2022_graph, N2O_2023_graph,
  nrow = 3,
  common.legend = TRUE,
  legend= "bottom")

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
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## Warning in is.na(x): is.na() applied to non-(list or vector) of type
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## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

## Warning in is.na(x): is.na() applied to non-(list or vector) of type
## 'expression'

ggsave(all_years_N2O, filename = "all_years_N2O.jpg", height = 25, width = 25, units = "cm", dpi=400)
```